



TrendAgent PoC

AgenticAI powered by Gemini for Instant Analysis



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Annexure: I – II



Problem Statement:

- Automate data cleaning, mapping, and merging
- Multiple file format inputs (.csv, .xlsx), with inconsistent headers, and types
- Run multi-question and cross-dataset analysis
- Run a suite of classic tests on datasets based on agent suggestions

User Stories



Cross-Dataset Analysis

- I want to select variables from multiple datasets so I can discover trends and relationships that span across sources.

Data Exploration

- I want suggestions for EDA so I know which distributions, outliers, or patterns to focus on.

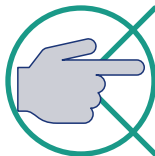
AI-Guided Tests

- I want to know what the most informative tests and charts for my selected variables.

Freeform Support

- I want to ask open-ended questions about my data and findings so I can interpret results and decide on next steps.

What Makes TrendAgent Unique



Select columns from any uploaded file, compare or trend them—no need to pre-merge



Smart key detection for row-level merges and joint analysis



AI-driven recommendations for crosstabs, segmentation, and variable relationships



Effortless exploration of trends, group differences, and correlations that span datasets

Gemini Throughout the Workflow



Proof of Concept Workflow

Deployed PoC: [TrendAgent](#)

Upload Data

- Drag-and-drop multiple files, any format (.csv, .xlsx, tab-separated)

Cross-Dataset Selection

- Choose any columns, instantly see what can be compared or merged

Crosstab & Trend Analysis

- Run group trends and tests with AI suggestions
- Visualize fully customized insights with a click

EDA & Profiling

- View column profiles and summaries
- Get instant customized data exploration results

Custom LLM's

- Ask for trend explanations, and custom test insights
- Get custom next step analysis roadmaps with Gemini

Export Results

- Download all outputs for fast sharing and reporting

Layers of Intelligence & Automation



Harmonization: Auto-aligns columns, formats, and types across files

Validation & Profiling: AI detects missingness, uniqueness, and flags issues

Integrated Analytics: Cross-dataset trends, crosstabs, group insights, and correlation analysis

Custom LLM: Powers adaptive, context-aware analysis and recommendations

Narrative Output: Every chart and table is supported by Gemini-powered explanations.



Thank you

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Annexure I – Workflow Visuals

Upload datasets

Upload Datasets

Upload CSV or Excel

Drag and drop files here
Limit 200MB per file • CSV, XLSX

Browse files

market_sales_market1.csv
0.6MB

consumer_survey.csv
291.7KB

Tendency AI Vision:
AI for multi-survey trend detection, scenario simulation, group insights, and instant report generation.

Data Dictionary

TrendAgent

AI Agent for Instant EDA, Trend Flagging, and Group Insights

Data Dictionary Cross-Dataset Trends & Groups EDA AI Features & Anomalies (Coming Soon) Report Generation (Coming Soon) External

Data Dictionary

What does this do?

consumer_survey.csv

	Column	Type	Unique	Missing	Sample
0	RespondentID	int64	10000	0	1
1	Region	object	4	0	Northeast
2	ProductSatisfaction	int64	5	0	3
3	BrandLoyalty	int64	2	0	0
4	RecommendationLikely	int64	2	0	1
5	CustomerAge	int64	84	0	33
6	IncomeBracket	object	3	0	High
7	LastPurchaseMonthsAgo	int64	22	0	4

Annexure I – Workflow Visuals

Dictionary LLM

Gemini AI Q&A: Ask about your schema or next steps.

Ask Gemini (dictionary tab)

from the dictionary can you suggest what trends should i look for in the cross data analysis and eda

Ask on Dictionary

To perform a cross-data analysis and EDA combining `consumer_survey.csv` and `market_sales_market1.csv`, focusing on identifying trends, consider these avenues:

I. Linking the Datasets:

The key to combining these datasets is the shared `Region` and `CustomerAge` fields. You'll likely need to aggregate data before joining, especially for `CustomerAge` which may be grouped differently (e.g., ranges in the survey vs. exact ages in sales data). Consider these steps:

- Age Binning:** Harmonize `CustomerAge`. Decide on appropriate age bins (e.g., 18-25, 26-35, etc.) and create new columns in both datasets representing these bins.
- Joining:** Use the harmonized `CustomerAge` bins and `Region` to perform a join (likely a left join using the survey as the base, unless you have reasons to prioritize sales data).

II. Trends to Explore:

After joining, analyze the following, considering potential interactions between variables:

A. Regional Differences:

- Product Satisfaction & Sales:** Does higher product satisfaction in a region correlate with higher unit sales or higher average transaction values ($\text{UnitsSold} * \text{UnitPrice} * (1 - \text{DiscountRate})$) in that region? Are there regions where certain product categories underperform despite high satisfaction?
- Brand Loyalty & Sales:** Does higher brand loyalty predict repeat purchases (indicated by `LastPurchaseMonthsAgo` and `transaction`)

Manage app

Cross Dataset Analysis

TrendAgent

AI Agent for Instant EDA, Trend Flagging, and Group Insights

Data Dictionary Cross-Dataset Trends & Groups EDA AI Features & Anomalies (Coming Soon) Report Generation (Coming Soon) External Data Integration (Coming Soon)

Cross-Dataset Trend Analysis & Dashboard Graphs

How does this work?

1. Select datasets/columns for analysis

How many datasets to analyze?

2

Dataset #1

market_sales_market1.csv

Dataset #2

consumer_survey.csv

Columns from market_sales_market1.csv

CustomerAge x DiscountRate x UnitPrice x UnitsSold

Columns from consumer_survey.csv

IncomeBracket x CustomerAge x Recommendation... x BrandLoyalty

Suggested merge possible on columns: CustomerAge, Region

☐ Merge datasets on a common key for row-level analysis?

Manage app

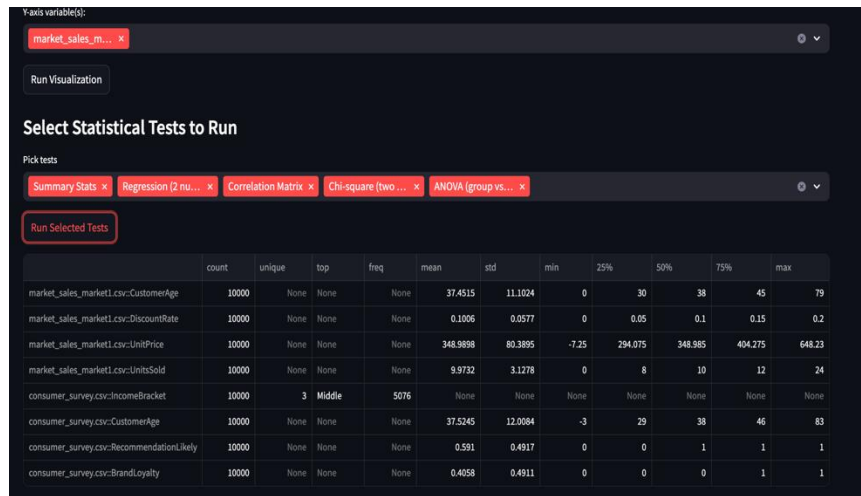
Option to merge dataset

Annexure I – Workflow Visuals

Cross dataset viz

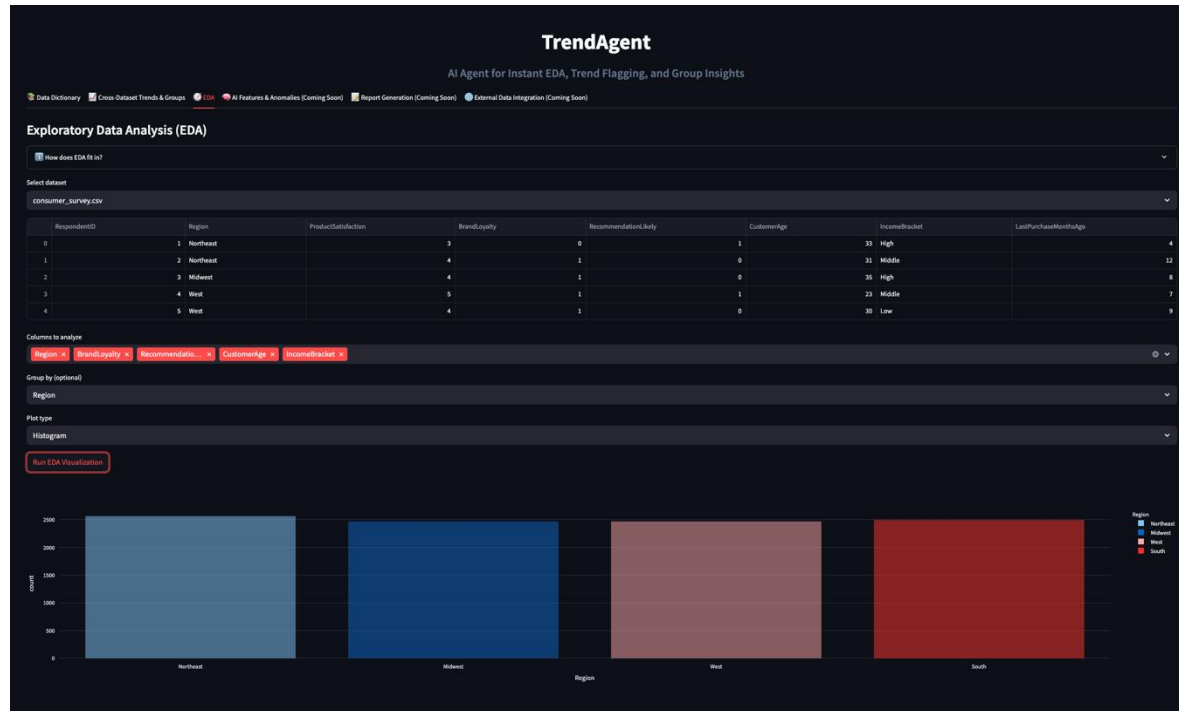


Cross dataset tests



Annexure I – Workflow Visuals

EDA tab



Annexure I – Workflow Visuals



Custom LLM – Cross-data set and EDA

Gemini AI: Insight from Cross-Dataset Analysis

Ask Gemini (Cross-Analysis)

Ask Gemini about Cross-Analysis

Gemini AI: Exploratory Data Analysis Ideas

Ask Gemini (EDA)

Ask Gemini about EDA

Free form Gemini

Freeform Gemini Q&A

Ask Gemini anything about your data, analysis, or trends.

Ask Gemini

Annexure II – Roadmap



Automated Anomaly & Pattern Detection

Instantly flag and explain outliers, emerging trends, or data quality issues across datasets.

Automated Dashboard Generation (PowerBI)

One-click reports—charts, tables, and narrative insights ready

Real-Time Data Integration

Sync and analyze live data streams, including:
Google Sheets and Workspace

APIs for economic indicators and external benchmarks

News feeds, public data repositories, and proprietary data sources

Annexure II – Roadmap



Continuous Learning AI Engine

Model adapts to new data, feedback, and use cases—improving recommendations and analysis over time.

Advanced Scenario Simulation

AI-powered “what-if” tools to forecast impact and model outcomes under changing conditions.

Custom Dashboards & Visualization Templates

Interactive, user-configurable dashboards tailored for different teams and reporting needs.

Expanded API Ecosystem

Integration with additional platforms (CRM, ERP, third-party analytics) for enterprise-scale workflows.